

Repeated Exam MA-224

Course code: MA-224

Course name: Discrete mathematics 2

Date: 18. February 2025

Time: 09:00 – 13:00 (4 hours)

Number of pages: 5

Number of tasks: 12

Permitted equipment: All written or printed material. Calculator.
No communication is allowed during the exam.

Remarks: Present your answers so that it is clear what methods you have used, and how you have used them. Your answer should contain a calculation or an argument explaining how the answer was reached; otherwise, it is incomplete.
All problems 1-12 are scored equally. If a problem has more than one part, then each part is scored equally within the problem.

Problem 1

We consider the following theorem.

$$\text{Th1: } [(r \rightarrow p) \vee (\neg q \rightarrow \neg r)] \iff [(r \wedge \neg q) \rightarrow p]$$

- (a) Prove Th1 using truth tables.
- (b) Prove Th1 using logical inferences.

For your convenience, here are some possible inference rules.

equivalence between $\rightarrow$ and $\wedge$: $A \rightarrow B \iff \neg A \vee B$

double negation for $\neg$: $\neg\neg A \iff A$

idempotence law for $\wedge$: $A \wedge A \iff A$

idempotence law for $\vee$: $A \vee A \iff A$

deMorgan's Law for $\wedge$: $\neg(A \wedge B) \iff \neg A \vee \neg B$

deMorgan's Law for $\vee$: $\neg(A \vee B) \iff \neg A \wedge \neg B$

commutativity of $\wedge$: $A \wedge B \iff B \wedge A$

associativity of $\wedge$: $(A \wedge B) \wedge C \iff A \wedge (B \wedge C)$

distributivity of $\wedge$: $(A \vee B) \wedge C \iff (A \wedge C) \vee (B \wedge C)$

commutativity of $\vee$: $A \vee B \iff B \vee A$

associativity of $\vee$: $(A \vee B) \vee C \iff A \vee (B \vee C)$

distributivity of $\vee$: $(A \wedge B) \vee C \iff (A \vee C) \wedge (B \vee C)$

Problem 2

Define the following sequence recursively: $s_i = i^2 + i + 1$. Give reasons.

Problem 3

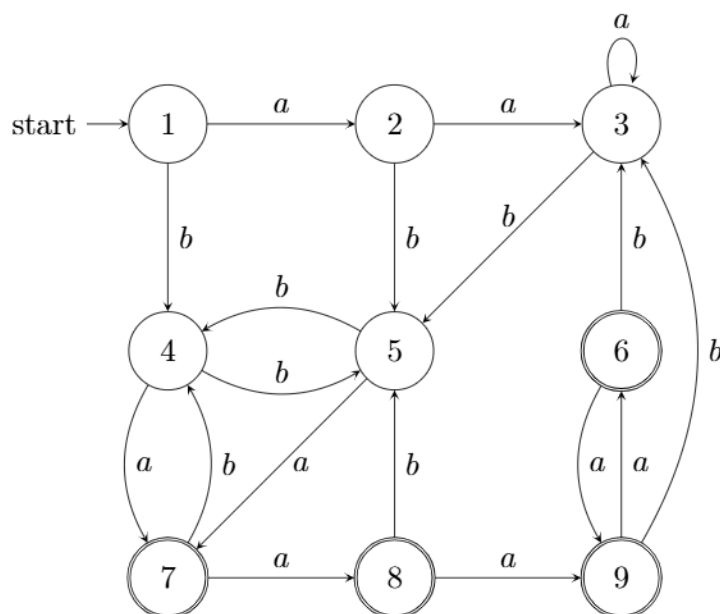
Consider the following EBNF grammar with the start symbol s . Terminals are enclosed in single quotes ('jump'). Alternatives are denoted by '|'.

$$s \rightarrow \text{'jump'} \mid \text{'step'} \mid \text{'before'} s \text{'do'} s \mid s \text{'and'} s$$

- (a) Find a derivation in the EBNF grammar above for the string *before step do jump and step*.
- (b) Is the grammar ambiguous? Give reasons.

Problem 4

Minimize the following DFA. Show all relevant steps.



Problem 5

Is the code below correct related to the following specification?

Specification: $\text{SQRT}(x, \text{res}:\text{NUM}) \text{ res} := \text{res}' \mid \text{res}' * \text{res}' = x$

Code: `float sqrt(x:float) { return (-1) * Math.sqrt(x); }`

Problem 6

Let R be a symmetric and transitive relation of natural numbers. Each natural number is related with R to at least one natural number.

Prove that R is an equivalence relation or provide an example where this is not the case.

Problem 7

Find the solution to the initial value problem

$$\begin{cases} a_n - 4a_{n-1} - 4a_{n-2} = 0 \\ a_0 = 0 \\ a_1 = 1. \end{cases}$$

Problem 8

Let U be the set of 2×2 matrices:

$$U = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R}, \text{ such that } a \cdot c \neq 0 \right\}.$$

Prove that $(U, \cdot)$ is a group when the group operation $\cdot$ is matrix multiplication.

When solving this problem, it helps to know that the inverse of a general 2×2 -matrix can be computed by the formula

$$\begin{bmatrix} x & y \\ z & w \end{bmatrix}^{-1} = \frac{1}{xw - yz} \begin{bmatrix} w & -y \\ -z & x \end{bmatrix},$$

if it exists.

Problem 9

Let $(P_{15}, \cdot_{15})$ be the group where

$$P_{15} = \{1 \leq x \leq 15 \mid \gcd(x, 15) = 1\},$$

and the group operation is integer multiplication modulo 15.

The following is the multiplication table for P_{15} where a few entries are missing.

$\cdot_{15}$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	1	7	11	13
4	4	8	-	-	2	14	7	11
7	7	14	13	4	11	2	-	8
8	8	1	2	-	4	13	14	7
11	11	7	14	2	13	1	8	4
13	13	11	7	1	14	8	-	2
14	14	13	11	8	7	4	2	1

(a) Calculate the missing entries in the multiplication table:

$$4 \cdot_{15} 4 =$$

$$4 \cdot_{15} 7 =$$

$$8 \cdot_{15} 7 =$$

$$7 \cdot_{15} 13 =$$

$$13 \cdot_{15} 13 =$$

(b) Is P_{15} an Abelian group? Prove your answer.

(c) Is P_{15} a cyclic group? Prove your answer.

Problem 10

Let $(\mathbb{Z}/n, +_n)$ be the group of integers modulo n where the group operation is addition modulo n . Let $n = 211$. Prove that the element $10 \in \mathbb{Z}/211$ is a generator.

Problem 11

Let the encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be given by multiplication with the matrix:

$$E(w_1 \ w_2 \ w_3) = (w_1 \ w_2 \ w_3) \cdot \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

The message x is encoded using E and transmitted over a noisy channel. The bitstring $r = (00110)$ is received. We assume that the transmission error weight is at most 1 during this transmission.

(a) Prove that r is a code word and find the original message x .

(b) Prove that E is able to detect all transmission errors of weight 1.

Problem 12

Given the (3×6) generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Assume that a 3-bit message is encoded using G and transmitted over a noisy channel, and that the received message is $r = (101100)$. Assume that at most one bit error occurs during the transmission of the message.

Compute the syndrome of r and use it to detect and correct any error in the transmission.